

Step 1: PCA diagnostics for choosing m**The FACTOR Procedure**

Input Data Type	Raw Data
Number of Records Read	572
Number of Records Used	572
N for Significance Tests	572

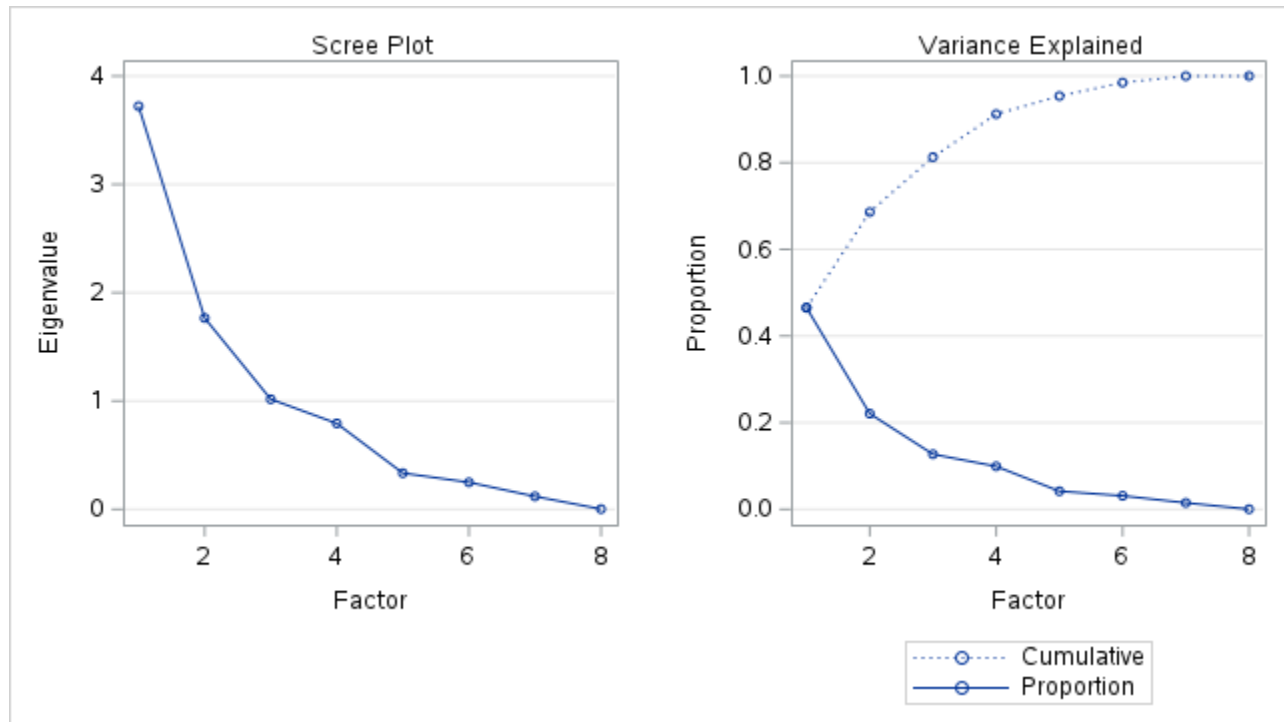
Step 1: PCA diagnostics for choosing m**The FACTOR Procedure**

Initial Factor Method: Principal Components

Prior Communality Estimates: ONE

Eigenvalues of the Correlation Matrix: Total = 8 Average = 1				
	Eigenvalue	Difference	Proportion	Cumulative
1	3.72141001	1.95561249	0.4652	0.4652
2	1.76579752	0.74944209	0.2207	0.6859
3	1.01635543	0.22345660	0.1270	0.8129
4	0.79289883	0.45908117	0.0991	0.9121
5	0.33381767	0.08499900	0.0417	0.9538
6	0.24881867	0.12999856	0.0311	0.9849
7	0.11882011	0.11673835	0.0149	0.9997
8	0.00208176		0.0003	1.0000

3 factors will be retained by the MINEIGEN criterion.



Factor Pattern			
	Factor1	Factor2	Factor3
palmitic	0.88882	-0.06589	0.11539
palmitoleic	0.86853	-0.32013	0.14376
stearic	-0.19030	0.34334	0.80869

Factor Pattern			
	Factor1	Factor2	Factor3
oleic	-0.95331	0.21083	-0.08077
linoleic	0.70546	-0.45632	-0.08819
linolenic	0.42245	0.80373	-0.19259
arachidic	0.44042	0.59425	-0.43012
eicosenoic	0.60162	0.53787	0.30331

Variance Explained by Each Factor		
Factor1	Factor2	Factor3
3.7214100	1.7657975	1.0163554

Final Community Estimates: Total = 6.503563							
palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
0.80765430	0.87749037	0.80807910	0.95977611	0.71368157	0.86153110	0.73210088	0.74324954

Step 2: PCA m=3, varimax rotation

The FACTOR Procedure

Input Data Type	Raw Data
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Correlations								
	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
palmitic	1.00000	0.83560	-0.17039	-0.83734	0.46068	0.31933	0.22830	0.50195
palmitoleic	0.83560	1.00000	-0.22219	-0.85244	0.62163	0.09311	0.08548	0.41635
stearic	-0.17039	-0.22219	1.00000	0.11360	-0.19782	0.01892	-0.04098	0.14038
oleic	-0.83734	-0.85244	0.11360	1.00000	-0.85032	-0.21817	-0.31996	-0.42415
linoleic	0.46068	0.62163	-0.19782	-0.85032	1.00000	-0.05744	0.21097	0.08904
linolenic	0.31933	0.09311	0.01892	-0.21817	-0.05744	1.00000	0.62024	0.57832
arachidic	0.22830	0.08548	-0.04098	-0.31996	0.21097	0.62024	1.00000	0.32866
eicosenoic	0.50195	0.41635	0.14038	-0.42415	0.08904	0.57832	0.32866	1.00000

Step 2: PCA m=3, varimax rotation

The FACTOR Procedure Initial Factor Method: Principal Components

Partial Correlations Controlling all other Variables								
	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
palmitic	1.00000	-0.49371	-0.81558	-0.96675	-0.96050	-0.30224	-0.51243	-0.50208
palmitoleic	-0.49371	1.00000	-0.62247	-0.65857	-0.60200	-0.38686	-0.46754	-0.16313
stearic	-0.81558	-0.62247	1.00000	-0.84270	-0.83486	-0.36214	-0.47383	-0.31466
oleic	-0.96675	-0.65857	-0.84270	1.00000	-0.99043	-0.37359	-0.54207	-0.53022
linoleic	-0.96050	-0.60200	-0.83486	-0.99043	1.00000	-0.39491	-0.49354	-0.54421
linolenic	-0.30224	-0.38686	-0.36214	-0.37359	-0.39491	1.00000	0.22546	0.15585
arachidic	-0.51243	-0.46754	-0.47383	-0.54207	-0.49354	0.22546	1.00000	-0.28058

Partial Correlations Controlling all other Variables								
	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
eicosenoic	-0.50208	-0.16313	-0.31466	-0.53022	-0.54421	0.15585	-0.28058	1.00000

Kaiser's Measure of Sampling Adequacy: Overall MSA = 0.36626306							
palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
0.37770058	0.52941582	0.04962441	0.39785809	0.27911482	0.54256492	0.33795395	0.50405344

Step 2: PCA m=3, varimax rotation

The FACTOR Procedure
Initial Factor Method: Principal Components

Prior Communality Estimates: ONE

Eigenvalues of the Correlation Matrix: Total = 8 Average = 1				
	Eigenvalue	Difference	Proportion	Cumulative
1	3.72141001	1.95561249	0.4652	0.4652
2	1.76579752	0.74944209	0.2207	0.6859
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6	0.24881867	0.12999856	0.0311	0.9849
7	0.11882011	0.11673835	0.0149	0.9997
8	0.00208176		0.0003	1.0000

3 factors will be retained by the NFACTOR criterion.

Factor Pattern			
	Factor1	Factor2	Factor3
palmitic	0.88882	-0.06589	0.11539
palmitoleic	0.86853	-0.32013	0.14376
stearic	-0.19030	0.34334	0.80869
oleic	-0.95331	0.21083	-0.08077
linoleic	0.70546	-0.45632	-0.08819
linolenic	0.42245	0.80373	-0.19259
arachidic	0.44042	0.59425	-0.43012
eicosenoic	0.60162	0.53787	0.30331

Variance Explained by Each Factor		
Factor1	Factor2	Factor3
3.7214100	1.7657975	1.0163554

Final Communality Estimates: Total = 6.503563							
palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
0.80765430	0.87749037	0.80807910	0.95977611	0.71368157	0.86153110	0.73210088	0.74324954

Residual Correlations With Uniqueness on the Diagonal								
	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
palmitic	0.19235	0.02596	-0.07195	0.03320	-0.18623	0.01903	-0.07437	-0.03234
palmitoleic	0.02596	0.12251	-0.06326	0.05464	-0.12449	0.01119	-0.04497	0.02241
stearic	-0.07195	-0.06326	0.19192	-0.07488	0.16442	-0.02090	0.18664	-0.17509

Residual Correlations With Uniqueness on the Diagonal								
	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
oleic	0.03320	0.05464	-0.07488	0.04022	-0.08871	-0.00046	-0.06013	0.06048
linoleic	-0.18623	-0.12449	0.16442	-0.08871	0.28632	-0.00569	0.13351	-0.06319
linolenic	0.01903	0.01119	-0.02090	-0.00046	-0.00569	0.13847	-0.12627	-0.04972
arachidic	-0.07437	-0.04497	0.18664	-0.06013	0.13351	-0.12627	0.26790	-0.12547
eicosenoic	-0.03234	0.02241	-0.17509	0.06048	-0.06319	-0.04972	-0.12547	0.25675

Root Mean Square Off-Diagonal Residuals: Overall = 0.09324089								
palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic	
0.08330099	0.06071917	0.12406607	0.05956906	0.12359660	0.05260685	0.11681030	0.09108200	

Partial Correlations Controlling Factors								
	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
palmitic	1.00000	0.16911	-0.37446	0.37740	-0.79358	0.11659	-0.32761	-0.14553
palmitoleic	0.16911	1.00000	-0.41254	0.77843	-0.66469	0.08588	-0.24821	0.12633
stearic	-0.37446	-0.41254	1.00000	-0.85229	0.70141	-0.12822	0.82309	-0.78877
oleic	0.37740	0.77843	-0.85229	1.00000	-0.82661	-0.00612	-0.57928	0.59516
linoleic	-0.79358	-0.66469	0.70141	-0.82661	1.00000	-0.02855	0.48206	-0.23305
linolenic	0.11659	0.08588	-0.12822	-0.00612	-0.02855	1.00000	-0.65558	-0.26371
arachidic	-0.32761	-0.24821	0.82309	-0.57928	0.48206	-0.65558	1.00000	-0.47842
eicosenoic	-0.14553	0.12633	-0.78877	0.59516	-0.23305	-0.26371	-0.47842	1.00000

Root Mean Square Off-Diagonal Partial: Overall = 0.51092638								
palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic	
0.39335955	0.43613355	0.63750131	0.63784021	0.60170524	0.27712696	0.54419386	0.44178993	

Step 2: PCA m=3, varimax rotation

The FACTOR Procedure
Rotation Method: Varimax

Orthogonal Transformation Matrix			
	1	2	3
1	0.90612	0.42194	-0.03010
2	-0.38372	0.84983	0.36130
3	0.17803	-0.31583	0.93196

Rotated Factor Pattern			
	Factor1	Factor2	Factor3
palmitic	0.85121	0.28259	0.05698
palmitoleic	0.93543	0.04901	-0.00782
stearic	-0.16021	-0.04392	0.88345
oleic	-0.95910	-0.19755	0.02960
linoleic	0.79864	-0.06228	-0.26829
linolenic	0.04009	0.92211	0.09818
arachidic	0.09447	0.82669	-0.19941
eicosenoic	0.39275	0.61516	0.45889

Variance Explained by Each Factor

Factor1	Factor2	Factor3
3.3477233	2.0392029	1.1166368

Final Community Estimates: Total = 6.503563							
palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
0.80765430	0.87749037	0.80807910	0.95977611	0.71368157	0.86153110	0.73210088	0.74324954

Step 2: PCA m=3, varimax rotation

The FACTOR Procedure
Rotation Method: Varimax

